



# **Data Center Management**

# Data Center Management



- Data center management refers to the set of tasks and activities handled by an organization for the day-to-day requirements of operating a data center
- Data centers are increasingly complex facilities to operate, involving the management of hardware, software, services and physical infrastructure
- On the hardware side, data centers include many types of equipment, including compute and storage hardware that needs to be managed for ongoing operations
- Software and virtual workloads that run on racks of server hardware in a data center are often also within the purview of data center management operations
- Managing the physical infrastructure of a data center is another primary task within the domain of data center management
- The physical infrastructure includes the power and cooling required for regular operations, as well as backup systems for disaster recovery, power interruptions, and security and access control to the data center premises

# Challenges of data center management



- **Capacity management**

- Managing the available resources to meet user demands is always a challenge

- **Service-level agreements**

- Beyond just managing capacity, data centers need to operate at specific levels of availability and performance to comply with SLAs

- **Change management**

- There's a need to constantly iterate on hardware and software within a data center
- Managing the changes, and the processes that enable change, can be a challenge

- **Disaster recovery**

- Businesses should prepare their data centers for unexpected events, such as a power failures, cyber attacks or hardware issues



# Challenges of data center management

- **Real-time reporting**

- The ability to provide real-time data on the state of all data center operations at any given point in time is often a complex issue

- **Security**

- Physical security for data center assets and IT security for running workloads in a multi-tenant data center can be complicated

- **Power management**

- Managing the power utilization of data center operations is an ongoing challenge for many operators who aim to balance power consumption and cost concerns with the need to maintain SLAs for running workloads



# Best practices for data center management

- **Measure everything**

- You can't accurately manage anything without first measuring it
- Having data measurement and logs for all assets in a data center is important

- **Understand effective power usage**

- It is among the most critical metrics for data center management because it helps operators to understand how efficiently power is used

- **Consider cooling options**

- Cooling is one of the most power-intensive aspects of data center management
- Using optimized air flow management techniques and ambient cooler air in certain climates can help to lower costs



# Best practices for data center management

- **Backup and more backups**
  - Redundant systems and backups for power and operational systems is an important best practice
- **Predictive and proactive maintenance**
  - Waiting until a given data center asset fails before replacing it is a risky proposition that can affect availability
  - Predictive and proactive maintenance and replacement policies help fix and restore assets before they fail
- **Embrace DCIM**
  - For a successful DCIM implementation, organizations need to be aware of and have the right deployment DCIM technology to help them optimize data center processes and operations

# Components of a data center



## ■ Facility

- The facility is the physical building used for the data center
- In simplest terms, a data center is just a big open space where infrastructure will be deployed
- Although almost any space has the potential to operate some amount of IT infrastructure, a properly designed facility considers multiple array of factors

## ■ Space

- There must be sufficient floor space -- a simple measure of square feet or square meters -- to hold all the IT infrastructure that the business intends to deploy now and in the future'
- The space must be located on a well-considered site with affordable taxes and access
- The space is often subdivided to accommodate different purposes or use types

## ■ Power

- There must be adequate power -- in watts, often as much as 100 megawatts -- to operate all the IT infrastructure
- Power must be affordable, clean -- meaning free of fluctuation or disruption -- and reliable
- Renewable and supplemental/auxiliary power must be included

# Components of a data center



## ■ Cooling

- The enormous amount of power delivered to a data center is converted into computing -- i.e., work -- and a lot of heat, which must be removed from the IT infrastructure using conventional HVAC systems, as well as other unconventional cooling technologies

## ■ Security

- Considering the value of the data center and its critical importance to the business, the data center must include controlled access using a variety of tactics, ranging from employee badge access to video surveillance

## ■ Management

- Modern data centers typically incorporate a building management system (BMS) designed to help IT and business leaders oversee the data center environment in real time, including oversight of temperature, humidity, power and cooling levels, as well as access and security logging



# Components of a data center



- **Enterprise data storage**

- A modern data center houses an organization's data systems in a well-protected physical and storage infrastructure along with servers, storage subsystems, networking switches, routers, firewalls, cabling and physical racks

- **Support infrastructure**

- This equipment provides the highest available sustainability related to uptime
- Components of the support infrastructure include the following:
  - power distribution and supplemental power subsystems
  - electrical switching
  - UPSes
  - backup generators
  - ventilation and data center cooling systems, such as in-row cooling configurations and computer room air conditioners
  - adequate provisioning for network carrier, or telecom, connectivity

- **Operational staff**

- These employees are required to maintain and monitor IT and infrastructure equipment around the clock

# Types of Data Centers



- **Enterprise data centers**

- These proprietary data centers are built and owned by organizations for their internal end users
- They support the IT operations and critical applications of a single organization and can be located both on-site and off-site

- **Managed services data centers**

- Managed by third parties, these data centers provide all aspects of data storage and computing services
- Companies lease, instead of buy, the infrastructure and services

- **Cloud-based data centers**

- These off-site distributed data centers are managed by third-party or public cloud providers, such as Amazon Web Services, Microsoft Azure or Google Cloud. Based on an infrastructure-as-a-service model, the leased infrastructure enables customers to provision a virtual data center within minutes

- **Edge data centers**

- These are smaller facilities that solve the latency problem by being geographically closer to the edge of the network and data sources

# Types of Data Centers



## ■ Colocation data centers

- These rental spaces inside colocation facilities are owned by third parties
- The renting organization provides the hardware, and the data center provides and manages the infrastructure, including physical space, bandwidth, cooling and security systems
- Colocation is appealing to organizations that want to avoid the large capital expenditures associated with building and maintaining their own data centers

## ■ Hyperscale data centers

- Synonymous with large-scale providers, such as Amazon, Meta and Google, these hyperscale computing infrastructures maximize hardware density, while minimizing the cost of cooling and administrative overhead

# Hardware Requirements



## ■ Servers

- Servers are the heart of a data center
- These units of hardware are responsible for storage, analysis and transmission of data
- The three major types of servers are blade servers, rack servers and tower servers
- Modern data centers use rack servers, wherein multiple servers are stacked within a rack

## ■ Racks

- Selecting the right racks is an important consideration in the setup or management of a data center, and is also a factor that's often overlooked
- Server racks are of two major kinds – Open Racks and Cabinets – and support the physical weight of the servers, storage devices and switches
- While Open Racks allow for better visibility and easier handling, Cabinets are the more secure and stable units. It is important to select the right racks or cabinets for your requirements, in order to balance accessibility and functionality

## ■ Switches

- The connector units of data center hardware, switches are responsible for linking multiple devices together
- Switches function based on the hardware addresses of devices

# Hardware Requirements



## ■ Cables

- Coaxial, twisted pair and fiber optic cables are used in data centers to connect switches, storage devices and servers
- Fiber optic cables allow for faster transmission, and they are therefore used heavily in modern data centers
- As cables are the units ultimately responsible for the smooth and timely transmission of data across the network, they play a central role in the functioning of a data center
- The usage of fiber cable raceways helps to protect and segregate fiber cables, for increased efficiency

## ■ Storage

- NAS (Network Attached Storage) and SAN (Storage Area Network) are the two most commonly used types of storage systems in modern data centers, with SAN providing higher speeds and more efficient access to the stored files and block level data
- Apart from these basic units, without which the Data Center cannot function, many other factors also come into play which are sometimes not given the same importance by those of us setting up or engaging the services of a data center

## ■ Policies and Procedures

- Maintenance and management of the physical hardware requires the strict adherence to certain policies and procedures that need to be followed

# Hardware Requirements



## ■ Power Equipment

- Data centers require 24x7 power availability and must therefore employ backup generators and multiple data routes to ensure Uninterrupted Power Supply (UPS)
- This is true no matter the physical location of the center, but is especially the case when the data center is located in a country or area where power supply is known to be unreliable

## ■ Security Systems

- The physical space where the data center is set up must be resistant to natural disasters like flooding, earthquakes, fires, etc.
- This is of course in addition to physical security systems and access control systems to keep the data private

## ■ Cooling Systems

- Data centers require that multiple servers be placed in close proximity to one another, in a limited space
- That being the case, there are high chances of equipment overheating rapidly, in spite of the units' internal fans
- Server rooms must therefore be maintained at a specific temperature as prescribed and planned during setup, through the use of external HVAC equipment, for smooth functioning